Galaba Vamsi

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SUMMARY

Mechatronics student working where mechanisms, control and robot software meet: robot modeling in ROS 2 and MuJoCo, controllers carried down to embedded C, and evaluation of robot policies in simulation before they touch hardware.

EDUCATION

B.Tech. in Mechatronics, Indian Institute of Technology (IIT) Bhilai Aug 2023 – May 2027
Coursework: Optimal Control, Advanced Control Theory, Deep Learning for Computer Vision, Digital Image Processing, AI and ML

MECHATRONICS PROJECTS

Whiteboard Cleaning Robot (team project) 2025
ROS 2 Jazzy, Gazebo Harmonic, URDF/Xacro, ros2_control, rclpy

- Modeled a 3-axis Cartesian gantry (links, joints and actuation in URDF/Xacro) and drove it through ros2_control.
- Ran a raster cleaning path with 10 cm overlap over a 90 x 90 cm board in simulation.

Quarter-Car Active Suspension Control (Advanced Control Theory) 2026
MATLAB/Simulink, C, QEMU

- Reproduced an IEEE TSMC 2022 active-suspension controller within 6% and showed its gains fail a hold-out test.
- Found an 88 Hz mode that destabilizes the 50 Hz version; ported the controller to C for a Cortex-M4 (tested in QEMU).

Pneumatic Sorting Digital Twin 2026
Unity, C# | github.com/Galabavamsi/pneumatic-sorting-digital-twin

- Simulated 3 electropneumatic training kits (10k lines of C#): cylinders, valves and sensors behind PLC-style inputs and outputs, plus an operator HMI.
- Added fault injection (E-stop, low pressure, air leak, stuck actuator, failed sensor) and a 10 Hz telemetry log with replay.

SCARA Pick-and-Place Arm 2026
MuJoCo, Python

- Turns text commands (“sort red, blue, green”) into pick-and-place plans executed with analytic inverse kinematics.

Hardware builds

- Built a Mars rover for a NASA competition, an EEG-controlled wheelchair prototype and a sensor-based smart blind stick; started a rocketry club.

ROBOT SOFTWARE AND EVALUATION

Co-founder, Robotropic (physical-AI startup) | robotropic.com Oct 2026 – Present

- Building Harness to roll out new robot policies (VLA checkpoints) cell by cell, with shadow-run scorecards and rollback.
- Lead its factory simulator and change gateway, which checks every human or AI-agent action against safety limits.

Inspect Robots, Robocurve (YC S26): paper co-author; 2 merged PRs, 3 in review Sep 2026 – Oct 2026
Evals for LLM and VLA robot policies | Python | github.com/robocurve/inspect-robots

- Co-author of the paper (CoRL 2026 workshop); checkpoint/resume for long robot evals where safety aborts and hardware faults are never retried; 80 tests (#514, in review).

HFlow, Hebbian Robotics (YC S26): 8 merged PRs, 7 issues filed (all fixed) Sep 2026 – Oct 2026
Robot training-data SDK | Python, Apache Arrow, MCAP, LeRobot | github.com/Hebbian-Robotics/hflow

- Fixed robot-log export silently dropping fields (#657) and silent 12-bit depth loss on LeRobot import (#400).

World Model for Robot Manipulation: JEPA on LeRobot Push-T Aug 2026
PyTorch | write-up: galabavamsi.com/blog

- Trained a 1.9M-parameter world model on 160 Push-T episodes; latents captured the block's state, but planning scored 0. Published why.

SKILLS

Mechatronics: Robot modeling (URDF/Xacro), actuator and sensor integration, PLC-style I/O, embedded C (Cortex-M4)
Control: Inverse kinematics, LQR, Kalman filter, MATLAB/Simulink
Robot software: ROS 2 (rclpy, ros2_control), Gazebo, MuJoCo, Unity, LeRobot, PyTorch
Languages: Python, C, C++, Rust, C#

ACHIEVEMENTS

- Tech Lead, Inter IIT Tech Meet 14.0: 5th of 23 IITs (2025); Top 8 of 4,500+ teams, RAKSHAM AI Cybersecurity Hackathon (2026)
- Top 1% in JEE Advanced; Prime Minister's Scholarship Scheme scholar (2024)